

Lesion classes predict aphasia outcomes

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Abstract

Stroke recovery models can improve prognosis of therapy response in patients with chronic aphasia, yet quantifying the effect of lesion on recovery is challenging. This study evaluates lesion classification by gray matter alone and together with white matter, and which structural measures go with aphasia severity and naming.

Introduction

Aphasia is one of the most common and long-lasting sequelae of stroke, affecting approximately 30% of acute stroke survivors and persisting into the chronic stage in many of them. Treatment success was determined by calculating the proportion of potential maximal gain (PMG) for each trained category using the following formula: $\frac{(AVG\ Post - AVG\ Pre)}{(n\ items - AVG\ Pre)}$, where AVG Pre and AVG Post denote the averaged accuracy on trained items before and after treatment. Aphasia typically results in a constellation of language deficits that greatly impair communication abilities and decrease functional independence. Despite the chronicity of aphasia, patients can continue to regain lost language skills in the years beyond stroke onset with targeted treatment.

Variability in language recovery and response to treatment are hallmarks of the disorder. As such, predictive models that provide prescriptive treatment recommendations for patients according to demographic, stroke and deficit profiles are critical for chronic aphasia care, yet such models do not yet exist, and this paragraph runs on so that the page has enough running text in it to be measured.

Methods

Thirty-four individuals with chronic stroke-induced aphasia participated in the study. The primary eligibility requirement was presence of anomia. Exclusionary criteria included contraindications for MRI and active medical conditions that precluded study participation.

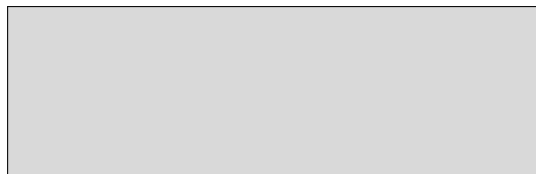


Fig. 1 Anatomical masks. Gray matter regions of interest, including DLPFC and iFrontal

Large gray matter masks were generated by combining regions from an atlas that aligned with major anatomical boundaries of the lateral surface of the brain. The rationale for using large regions was three-fold, which the reader can take on trust here.

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Compliance with ethical standards

Conflicts of interest This study was funded by the National Institutes of Health. The authors have no other financial conflicts of interest.

Table 1 Regions of interest included in large gray matter masks

Number	ROI mask	Atlas ROI
1	DLPFC	Superior frontal gyrus
2	DLPFC	Middle frontal gyrus
3	iFrontal	Inferior frontal gyrus
4	iFrontal	Insula

DLPFC dorsolateral prefrontal cortex

Table 2 Demographic and lesion data for all patients

ID	Sex	Hand	Age	MPO	Education	Lesion Volume (cc)	WAB-R AQ (/100)	PMG
P1	M	R	55	12	16	57.25	87.2	0.89
P2	F	L	50	29	16	249.93	25.2	0.00
P3	F	R	63	62	16	175.38	52.0	0.36

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